

Abdelrhman Abdelhamid Ahmed

Network Engineer | CCNP Enterprise | Azure | Windows Server | Network Automation with Python

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PROFILE

Network Engineer with Cisco Certified Network Professional (CCNP) Enterprise certification and Microsoft Certified: Azure Administrator Associate (AZ-104). Hands-on experience in enterprise networking, routing and switching, Windows Server administration, and Microsoft Azure. Skilled in OSPF, EIGRP, BGP, MPLS, VLANs, STP, Active Directory, and network troubleshooting. Experienced in network automation using Python and practical infrastructure labs involving Windows Server, virtualization, and Azure networking. Currently developing hands-on expertise in Microsoft Azure administration and cloud networking, with a strong interest in Network and Cloud Engineering.

PROFESSIONAL EXPERIENCE

May 2022 – May 2024
Hurghada, Egypt

Network Engineer

Hurghada Border Guard Intelligence Office, Egyptian Army

- Maintained and monitored network infrastructure supporting critical operational systems and reliable connectivity.
- Performed network troubleshooting to identify, diagnose, and resolve connectivity and infrastructure issues.
- Supported network availability and stable operations in a high-demand operational environment.
- Collaborated with team members to troubleshoot network issues and maintain reliable network connectivity.

ADDITIONAL EXPERIENCE (NON-NETWORKING ROLE)

Jun 2024 – Dec 2025
Aswan, Egypt

Safety Engineer

Enoia Utilities

- Reviewed work processes to identify safety risks and recommend corrective actions.
- Developed and implemented safety and quality control procedures while supporting inspections and compliance activities.

PROJECTS

Aug 2026 – Aug 2026

Azure Enterprise Network Lab | Microsoft Azure

- Designed and deployed an Azure network infrastructure using a Resource Group, Virtual Network (VNet), multiple Subnets, and Network Security Group (NSG).
- Deployed and configured Windows Server Virtual Machines with private/public IP addressing and IIS Web Server.
- Configured NSG inbound security rules to control RDP, HTTP, and Load Balancer traffic.
- Configured an Azure Load Balancer with a backend pool, HTTP health probe, and load-balancing rule to distribute web traffic across virtual machines.
- Deployed and configured an Azure VPN Gateway and established Point-to-Site (P2S) VPN connectivity for secure remote access.
- Documented the complete lab implementation and configuration steps in a GitHub-hosted PDF.

Jun 2026 – Jun 2026

Active Directory Infrastructure Lab | Windows Server 2025

- Designed and implemented a complete Active Directory environment using Windows Server 2025.
- Configured a Domain Controller, DNS, Organizational Units (OUs), and user accounts.
- Applied Group Policy Objects (GPOs) for security and user management.

Mar 2026 – Mar 2026

Network Path & OSPF Cost Simulator | Python

- Developed a Python-based tool to simulate OSPF path selection using Dijkstra's Algorithm.
- Applied Python scripting to model network topologies and simulate link-state path selection.
- Simulated routing scenarios to analyze path selection, routing behavior, and network optimization.

Oct 2021 – Jan 2022

Smart Museum System | IoT Graduation Project

- Contributed to an IoT-based museum monitoring system integrating metal detection, flame detection, and infrared (IR) sensing.
- Designed and integrated the metal detector, flame detector, and IR sensor components as part of the project.
- Implemented sensor-based monitoring and automated alerts to enhance artifact protection and visitor safety.

EDUCATION

Sep 2016 – Jan 2022
Cairo, Egypt

Bachelor of Engineering in Electronics and Communication Engineering

Modern Academy For Engineering & Technology | Cairo, Egypt

Sep 2016 – Jan 2022

- Graduation Project: Smart Museum — Grade: A+
- Overall Grade: Good

CERTIFICATIONS

- Cisco Certified Network Professional (CCNP) Enterprise — ENCSI (350-401)
- Cisco Certified Network Professional (CCNP) Enterprise — ENARSI (300-410)
- Cisco Certified Network Professional (CCNP) Enterprise — Combined Certification (ENCOR + ENARSI)
- Microsoft Certified: Azure Administrator Associate (AZ-104)

SKILLS

TECHNICAL SKILLS

Routing & Switching

- OSPF, EIGRP, BGP, IS-IS, MPLS
- VLANs, STP, VTP, DTP
- HSRP, VRRP, GLBP
- Network Design & Troubleshooting

Microsoft Azure / Cloud

- Azure Virtual Machines
- Azure Virtual Networks (VNet) & Subnets
- Network Security Groups (NSG)
- Azure Load Balancer
- Azure VPN Gateway & Point-to-Site (P2S) VPN
- Azure Networking
- Azure Resource Management

Windows Server & System Administration

- Windows Server 2016 / 2019 / 2022 / 2025
- Active Directory (AD)
- Group Policy (GPO)
- DNS & DHCP
- User & Access Management
- Basic Virtualization & Server Management

NETWORK AUTOMATION

- Python
- Network Automation & Scripting
- Network Path & OSPF Cost Simulation

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Communication
- Teamwork

COURSES

Jun 2026 – Aug 2026
Udemy

Microsoft Azure Administrator (AZ-104) — Certificate of Completion

- Microsoft Entra ID, Users & Group Management
- Azure Networking & Network Security Groups (NSG)
- Azure VPN & Azure Bastion
- Azure Storage Access Tiers & Redundancy
- Azure Resource Management

Jun 2026 – Aug 2026
Udemy

Azure Fundamentals (AZ-900) — Certificate of Completion

- Cloud Computing & Cloud Service Models
- Azure Architecture & Subscriptions
- Azure Virtual Machines & Virtual Networks
- Azure Cloud Concepts & Benefits

Apr 2026 – Jun 2026
Udemy

Microsoft Windows Server Administration 2025 — Certificate of Completion

- Windows Server Installation & Configuration
- Active Directory Domain Services (AD DS) & Group Policy (GPO)
- DNS & DHCP Configuration
- User, Group & Permission Management
- File Server, Backup & Recovery, and Remote Desktop Services (RDS)

Mar 2026 – Mar 2026
Online

Python Fundamentals

Codezilla channel

- Python basics: variables, data types, loops, and functions
- Basic scripting and automation tasks
- File handling and problem solving

Jun 2024 – Aug 2024
Online

CCNP Enterprise ENARSI Training

- Layer 2 protocols, MAC tables, CDP & LLDP
- VLANs, Trunks, DTP, VTP & Extended VLANs
- HSRP, VRRP, GLBP & Inter-VLAN Routing
- STP, RSTP, PVST+, Rapid-PVST & MSTP

Dec 2023 – Mar 2024
Online

CCNP Enterprise ENCOR Training

- OSPF, including single-area and multi-area configurations
- BGP, Route Maps & routing policies
- MPLS, MPLS Applications & IS-IS
- EIGRP, SD-WAN Architecture & Basic Routing Automation

Mar 2023 – May 2023
Online

CCNA Training

- Networking Fundamentals, Subnetting, Routing, Switching, Basic Configuration & IPv6

TRAINING

Jun 2026 – Aug 2026 Udemy	Microsoft Azure Administrator (AZ-104) — Certificate of Attendance — 80 Hours
Apr 2026 – Jun 2026 Udemy	Microsoft Windows Server Administration — Certificate of Attendance — 100 Hours
Feb 2022 – Mar 2022 Online	OSHA Construction Safety & Health — 30 Hours <i>Elsafwa Training Center</i>
Aug 2020 – Aug 2020 Online	Mobile Communications Fundamentals <i>Mega-Academy</i> • GSM, GPRS, EDGE, UMTS & CDMA
Jun 2019 – Sep 2019 Nasr City	Embedded Systems Fundamentals <i>IMT SCHOOL</i>
Aug 2018 – Aug 2018 Cairo, Egypt	Introduction To Programming of Arduino and Design implementation of electronic circuits <i>Modern Academy</i>
Jun 2017 – Jun 2017 Cairo, Egypt	Principles of Communication And Computing <i>Modern Academy</i>

LANGUAGES

Arabic: Native • English: Professional Working Proficiency